

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$\tan \theta + \cot \theta \equiv \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$	M1	2.1
	$\equiv \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$	A1	1.1b
	$\equiv \frac{1}{\frac{1}{2} \sin 2\theta}$	M1	2.1
	$\equiv 2 \operatorname{cosec} 2\theta \quad *$	A1*	1.1b
		(4)	
(b)	States $\tan \theta + \cot \theta = 1 \Rightarrow \sin 2\theta = 2$ AND no real solutions as $-1 \leq \sin 2\theta \leq 1$	B1	2.4
		(1)	
(5 marks)			

Notes:

(a)

M1: Writes $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and $\cot \theta = \frac{\cos \theta}{\sin \theta}$

A1: Achieves a correct intermediate answer of $\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$

M1: Uses the double angle formula $\sin 2\theta = 2 \sin \theta \cos \theta$

A1*: Completes proof with no errors. This is a given answer.

Note: There are many alternative methods. For example

$$\tan \theta + \cot \theta \equiv \tan \theta + \frac{1}{\tan \theta} \equiv \frac{\tan^2 \theta + 1}{\tan \theta} \equiv \frac{\sec^2 \theta}{\tan \theta} \equiv \frac{1}{\cos^2 \theta \times \frac{\sin \theta}{\cos \theta}} \equiv \frac{1}{\cos \theta \times \sin \theta} \text{ then as the}$$

main scheme.

(b)

B1: Scored for sight of $\sin 2\theta = 2$ and a reason as to why this equation has no real solutions.

Possible reasons could be $-1 \leq \sin 2\theta \leq 1$and therefore $\sin 2\theta \neq 2$

or $\sin 2\theta = 2 \Rightarrow 2\theta = \arcsin 2$ which has no answers as $-1 \leq \sin 2\theta \leq 1$

Q2.

Question	Scheme	Marks	AOs
(a)	$D = 5 + 2 \sin(30 \times 6.5)^\circ = \text{awrt } 4.48 \text{ m with units}$	B1	3.4
		(1)	
(b)	$3.8 = 5 + 2 \sin(30t)^\circ \Rightarrow \sin(30t)^\circ = -0.6$	M1	1.1b
		A1	1.1b
	$t = 10.77$	dM1	3.1a
	10:46 a.m. or 10:47 a.m.	A1	3.2a
		(4)	
(5 marks)			

Notes:

(a)

B1: Scored for using the model ie. substituting $t = 6.5$ into $D = 5 + 2 \sin(30t)^\circ$ and stating $D = \text{awrt } 4.48 \text{ m}$. The units must be seen somewhere in (a). So allow when $D = 4.482.. = 4.5 \text{ m}$. Allow the mark for a correct answer without any working.

(b)

M1: For using $D = 3.8$ and proceeding to $\sin(30t)^\circ = k$, $|k| \leq 1$

A1: $\sin(30t)^\circ = -0.6$ This may be implied by any correct answer for t such as $t = 7.2$

If the A1 implied, the calculation must be performed in degrees.

dM1: For finding the first value of t for their $\sin(30t)^\circ = k$ after $t = 8.5$.

You may well see other values as well which is not an issue for this dM mark

(Note that $\sin(30t)^\circ = -0.6 \Rightarrow 30t = 216.9^\circ$ as well but this gives $t = 7.2$)

For the correct $\sin(30t)^\circ = -0.6 \Rightarrow 30t = 323.1 \Rightarrow t = \text{awrt } 10.8$

For the incorrect $\sin(30t)^\circ = +0.6 \Rightarrow 30t = 396.9 \Rightarrow t = \text{awrt } 13.2$

So award this mark if you see $30t = \text{inv sin their } -0.6$ to give the first value of t where $30t > 255$

A1: Allow 10:46 a.m. (12 hour clock notation) or 10:46 (24 hour clock notation) oe

Allow 10:47 a.m. (12 hour clock notation) or 10:47 (24 hour clock notation) oe

DO NOT allow 646 minutes or 10 hours 46 minutes.

Q3.

Question	Scheme	Marks	AOs
(a)	$\cos 3A = \cos (2A + A) = \cos 2A \cos A - \sin 2A \sin A$	M1	3.1a
	$= (2 \cos^2 A - 1) \cos A - (2 \sin A \cos A) \sin A$	dM1	1.1b
	$= (2 \cos^2 A - 1) \cos A - 2 \cos A (1 - \cos^2 A)$	ddM1	2.1
	$= 4 \cos^3 A - 3 \cos A^*$	A1*	1.1b
		(4)	
(b)	$1 - \cos 3x = \sin^2 x \Rightarrow \cos^2 x + 3 \cos x - 4 \cos^3 x = 0$	M1	1.1b
	$\Rightarrow \cos x (4 \cos^2 x - \cos x - 3) = 0$ $\Rightarrow \cos x (4 \cos x + 3)(\cos x - 1) = 0$ $\Rightarrow \cos x = \dots$	dM1	3.1a
	Two of $-90^\circ, 0, 90^\circ$, awrt 139°	A1	1.1b
	All four of $-90^\circ, 0, 90^\circ$, awrt 139°	A1	2.1
		(4)	
(8 marks)			

(a)

Allow a proof in terms of x rather than A

M1: Attempts to use the compound angle formula for $\cos(2A + A)$ or $\cos(A + 2A)$

Condone a slip in sign

dM1: Uses correct double angle identities for $\cos 2A$ and $\sin 2A$

$\cos 2A = 2\cos^2 A - 1$ must be used. If either of the other two versions are used expect to see an attempt to replace $\sin^2 A$ by $1 - \cos^2 A$ at a later stage.

Depends on previous mark.

ddM1: Attempts to get all terms in terms of $\cos A$ using correct and appropriate identities.

Depends on both previous marks.

A1*: A completely correct and rigorous proof including correct notation, no mixed variables, missing brackets etc.

Alternative right to left is possible:

$$4\cos^3 A - 3\cos A = \cos A(4\cos^2 A - 3) = \cos A(2\cos^2 A - 1 + 2(1 - \sin^2 A) - 2) = \cos A(\cos 2A - 2\sin^2 A)$$

$$= \cos A \cos 2A - 2\sin A \cos A \sin A = \cos A \cos 2A - \sin 2A \sin A = \cos(2A + A) = \cos 3A$$

Score M1: For $4\cos^3 A - 3\cos A = \cos A(4\cos^2 A - 3)$

dM1: For $\cos A(2\cos^2 A - 1 + 2(1 - \sin^2 A) - 2)$ (Replaces $4\cos^2 A - 1$ by $2\cos^2 A - 1$ and $2(1 - \sin^2 A)$)

ddM1: Reaches $\cos A \cos 2A - \sin 2A \sin A$

A1: $\cos(2A + A) = \cos 3A$

(b)

M1: For an attempt to produce an equation just in $\cos x$ using both part (a) and the identity $\sin^2 x = 1 - \cos^2 x$

Allow one slip in sign or coefficient when copying the result from part (a)

dM1: Dependent upon the preceding mark. It is for taking the cubic equation in $\cos x$ and making a valid attempt to solve. This could include factorisation or division of a $\cos x$ term followed by an attempt to solve the 3 term quadratic equation in $\cos x$ to reach at least one non zero value for $\cos x$.

May also be scored for solving the cubic equation in $\cos x$ to reach at least one non zero value for $\cos x$.

A1: Two of $-90^\circ, 0, 90^\circ$, awrt 139° Depends on the first method mark.

A1: All four of $-90^\circ, 0, 90^\circ$, awrt 139° with no extra solutions offered within the range.

Note that this is an alternative approach for obtaining the cubic equation in (b):

$$1 - \cos 3x = \sin^2 x \Rightarrow 1 - \cos 3x = \frac{1}{2}(1 - \cos 2x)$$

$$\Rightarrow 2 - 2\cos 3x = 1 - \cos 2x$$

$$\Rightarrow 1 = 2\cos 3x - \cos 2x$$

$$\Rightarrow 1 = 2(4\cos^3 x - 3\cos x) - (2\cos^2 x - 1)$$

$$\Rightarrow 0 = 4\cos^3 x - 3\cos x - \cos^2 x$$

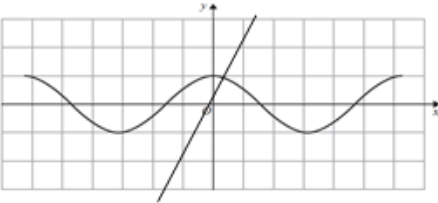
The M1 will be scored on the penultimate line when they use part (a) and use the correct identity for $\cos 2x$

Q4.

Question	Scheme	Marks	AOs
(a)	$R = \sqrt{109}$	B1	1.1b
	$\tan \alpha = \frac{3}{10}$	M1	1.1b
	$\alpha = 16.70^\circ$ so $\sqrt{109} \cos(\theta + 16.70^\circ)$	A1	1.1b
		(3)	
(b)	(i) e.g. $H = 11 - 10 \cos(80t)^\circ + 3 \sin(80t)^\circ$ or $H = 11 - \sqrt{109} \cos(80t + 16.70)^\circ$	B1	3.3
	(ii) $11 + \sqrt{109}$ or 21.44 m	B1ft	3.4
		(2)	
(c)	Sets $80t + "16.70" = 540$	M1	3.4
	$t = \frac{540 - "16.70"}{80} = (6.54)$	M1	1.1b
	$t = 6$ mins 32 seconds	A1	1.1b
		(3)	
(d)	Increase the '80' in the formula For example use $H = 11 - 10 \cos(90t)^\circ + 3 \sin(90t)^\circ$		3.3
		(1)	
(9 marks)			

Notes:
(a) B1: $R = \sqrt{109}$ Do not allow decimal equivalents M1: Allow for $\tan \alpha = \pm \frac{3}{10}$ A1: $\alpha = 16.70^\circ$
(b)(i) B1: see scheme (b)(ii) B1ft: their $11 +$ their $\sqrt{109}$ Allow decimals here.
(c) M1: Sets $80t + "16.70" = 540$. Follow through on their 16.70 M1: Solves their $80t + "16.70" = 540$ correctly to find t A1: $t = 6$ mins 32 seconds
(d) B1: States that to increase the speed of the wheel the 80's in the equation would need to be increased.

Q5.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)		M1	This mark is given for plotting the line $y = 2x + \frac{1}{2}$ on the diagram with a correct gradient and intercept
	Only one intersection means that there is one root	A1	This mark is given for a reason why there is only one real root
(b)	$1 - \frac{x^2}{2} - 2x - \frac{1}{2} = 0$	M1	This mark is given for using the small angle approximation $\cos x = 1 - \frac{x^2}{2}$ in the given equation
	$x^2 + 4x - 1 = 0$	M1	This mark is given for rearranging to find a quadratic equation to solve
	0.236 or $-2 + \sqrt{5}$	A1	This mark is given for finding the correct (positive) solution for x

Q6.

Part	Working or answer an examiner might expect to see	Mark	Notes
	$(10 \cos t)^2 + (4\sqrt{2} \sin t)^2 = 66$	M1	This mark is given for combining the two equations to show where the curve and circle meet
	$100 (\cos t)^2 + 32(1 - \cos t)^2 = 66$	M1	This mark is given for forming an equation in $\cos t$ only
	$68 \cos^2 t = 34$	A1	This mark is given for simplifying to find an equation in terms of $\cos t$
	$\cos t = \pm \frac{1}{\sqrt{2}} \Rightarrow t = \frac{\pi}{4}$	M1	This mark is given for finding a value for t
	$x = 10 \times \frac{1}{\sqrt{2}}$ $y = 4\sqrt{2} \times -\sin \frac{\pi}{4} = 4\sqrt{2} \times -\frac{1}{\sqrt{2}}$	M1	This mark is given for a method to substitute back into the original equations to find value for x and y
	$S = (5\sqrt{2}, -4)$	A1	This mark is given for the correct coordinates of S
(Total 6 marks)			

Q7.

Question	Scheme	Marks	AOs
	$1 - \cos 2\theta \equiv \tan \theta \sin 2\theta, \theta \neq \frac{(2n+1)\pi}{2}, n \in \mathbb{Z}$		
(a) Way 1	$\tan \theta \sin 2\theta = \left(\frac{\sin \theta}{\cos \theta}\right)(2 \sin \theta \cos \theta)$	M1	1.1b
	$= \left(\frac{\sin \theta}{\cancel{\cos \theta}}\right)(2 \sin \theta \cancel{\cos \theta}) = 2 \sin^2 \theta = 1 - \cos 2\theta *$	M1	1.1b
		A1*	2.1
		(3)	
(a) Way 2	$1 - \cos 2\theta = 1 - (1 - 2 \sin^2 \theta) = 2 \sin^2 \theta$	M1	1.1b
	$= \left(\frac{\sin \theta}{\cos \theta}\right)(2 \sin \theta \cos \theta) = \tan \theta \sin 2\theta *$	M1	1.1b
		A1*	2.1
		(3)	
	$(\sec^2 x - 5)(1 - \cos 2x) = 3 \tan^2 x \sin 2x, -\frac{\pi}{2} < x < \frac{\pi}{2}$		
(b) Way 1	$(\sec^2 x - 5) \tan x \sin 2x = 3 \tan^2 x \sin 2x$ or $(\sec^2 x - 5)(1 - \cos 2x) = 3 \tan x(1 - \cos 2x)$		
	Deduces $x = 0$	B1	2.2a
	Uses $\sec^2 x = 1 + \tan^2 x$ and cancels/factorises out $\tan x$ or $(1 - \cos 2x)$ e.g. $(1 + \tan^2 x - 3 \tan x - 5) \tan x = 0$ or $(1 + \tan^2 x - 3 \tan x - 5)(1 - \cos 2x) = 0$ or $1 + \tan^2 x - 5 = 3 \tan x$	M1	2.1
	$\tan^2 x - 3 \tan x - 4 = 0$	A1	1.1b
	$(\tan x - 4)(\tan x + 1) = 0 \Rightarrow \tan x = \dots$	M1	1.1b
	$x = -\frac{\pi}{4}, 1.326$	A1	1.1b
		A1	1.1b
		(6)	
(9 marks)			

Notes for Question	
(a)	Way 1
M1:	Applies $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and $\sin 2\theta = 2 \sin \theta \cos \theta$ to $\tan \theta \sin 2\theta$
M1:	Cancels as scheme (may be implied) and attempts to use $\cos 2\theta = 1 - 2 \sin^2 \theta$
A1*:	For a correct proof showing all steps of the argument
(a)	Way 2
M1:	For using $\cos 2\theta = 1 - 2 \sin^2 \theta$
Note:	If the form $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ or $\cos 2\theta = 2 \cos^2 \theta - 1$ is used, the mark cannot be awarded until $\cos^2 \theta$ has been replaced by $1 - \sin^2 \theta$
M1:	Attempts to write their $2 \sin^2 \theta$ in terms of $\tan \theta$ and $\sin 2\theta$ using $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and $\sin 2\theta = 2 \sin \theta \cos \theta$ within the given expression
A1*:	For a correct proof showing all steps of the argument
Note:	If a proof meets in the middle; e.g. they show $\text{LHS} = 2 \sin^2 \theta$ and $\text{RHS} = 2 \sin^2 \theta$; then some indication must be given that the proof is complete. E.g. $1 - \cos 2\theta \equiv \tan \theta \sin 2\theta$, QED, box

Notes for Question Continued			
(b)			
B1:	Deduces that the given equation yields a solution $x = 0$		
M1:	For using the key step of $\sec^2 x = 1 + \tan^2 x$ and cancels/factorises out $\tan x$ or $(1 - \cos 2x)$ or $\sin 2x$ to produce a quadratic factor or quadratic equation in just $\tan x$		
Note:	Allow the use of $\pm \sec^2 x = \pm 1 \pm \tan^2 x$ for M1		
A1:	Correct 3TQ in $\tan x$. E.g. $\tan^2 x - 3 \tan x - 4 = 0$		
Note:	E.g. $\tan^2 x - 4 = 3 \tan x$ or $\tan^2 x - 3 \tan x = 4$ are acceptable for A1		
M1:	For a correct method of solving their 3TQ in $\tan x$		
A1:	Any one of $-\frac{\pi}{4}$, awrt -0.785 , awrt 1.326 , -45° , awrt 75.964°		
A1:	Only $x = -\frac{\pi}{4}, 1.326$ cao stated in the range $-\frac{\pi}{2} < x < \frac{\pi}{2}$		
Note:	Alternative Method (Alt 1)		
	$(\sec^2 x - 5) \tan x \sin 2x = 3 \tan^2 x \sin 2x$ or $(\sec^2 x - 5)(1 - \cos 2x) = 3 \tan x(1 - \cos 2x)$		
	Deduces $x = 0$		B1 2.2a
	$\sec^2 x - 5 = 3 \tan x \Rightarrow \frac{1}{\cos^2 x} - 5 = 3 \left(\frac{\sin x}{\cos x} \right)$ $1 - 5 \cos^2 x = 3 \sin x \cos x$ $1 - 5 \left(\frac{1 + \cos 2x}{2} \right) = \frac{3}{2} \sin 2x$ $-\frac{3}{2} - \frac{5}{2} \cos 2x = \frac{3}{2} \sin 2x$ $\{3 \sin 2x + 5 \cos 2x = -3\}$	Complete process (as shown) of using the identities for $\sin 2x$ and $\cos 2x$ to proceed as far as $\pm A \pm B \cos 2x = \pm C \sin 2x$	M1 2.1
		$-\frac{3}{2} - \frac{5}{2} \cos 2x = \frac{3}{2} \sin 2x$ o.e.	A1 1.1b
	$\sqrt{34} \sin(2x + 1.03) = -3$	Expresses their answer in the form $R \sin(2x + \alpha) = k$; $k \neq 0$ with values for R and α	M1 1.1b
	$\sin(2x + 1.03) = -\frac{3}{\sqrt{34}}$		
	$x = -\frac{\pi}{4}, 1.326$		A1 1.1b
			A1 1.1b

Q8.

Question	Scheme	Marks	AOs
(a)	States or uses $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$	B1	1.2
	$\operatorname{cosec} \theta - \sin \theta = \frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta}$	M1	2.1
	$= \frac{\cos^2 \theta}{\sin \theta} = \cos \theta \times \frac{\cos \theta}{\sin \theta} = \cos \theta \cot \theta$ *	A1*	2.1
		(3)	
(b)	$\operatorname{cosec} x - \sin x = \cos x \cot(3x - 50^\circ)$ $\Rightarrow \cos x \cot x = \cos x \cot(3x - 50^\circ)$		
	$\cot x = \cot(3x - 50^\circ) \Rightarrow x = 3x - 50^\circ$	M1	3.1a
	$x = 25^\circ$	A1	1.1b
	Also $\cot x = \cot(3x - 50^\circ) \Rightarrow x + 180^\circ = 3x - 50^\circ$	M1	2.1
	$x = 115^\circ$	A1	1.1b
	Deduces $x = 90^\circ$	B1	2.2a
		(5)	
(8 marks)			
Notes:			

(a) Condone a full proof in x (or other variable) instead of θ 's here

B1: States or uses $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$ Do not accept $\operatorname{cosec} \theta = \frac{1}{\sin}$ with the θ missing

M1: For the key step in forming a single fraction/common denominator

E.g. $\operatorname{cosec} \theta - \sin \theta = \frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta}$. Allow if written separately $\frac{1}{\sin \theta} - \sin \theta = \frac{1}{\sin \theta} - \frac{\sin^2 \theta}{\sin \theta}$

Condone missing variables for this M mark

A1*: Shows careful work with all necessary steps shown leading to given answer. See scheme for necessary steps. There should not be any notational or bracketing errors.

(b) Condone θ 's instead of x 's here

M1: Uses part (a), cancels or factorises out the $\cos x$ term, to establish that one solution is found when

$$x = 3x - 50^\circ.$$

You may see solutions where $\cot A - \cot B = 0 \Rightarrow \cot(A - B) = 0$ or $\tan A - \tan B = 0 \Rightarrow \tan(A - B) = 0$.

As long as they don't state $\cot A - \cot B = \cot(A - B)$ or $\tan A - \tan B = \tan(A - B)$ this is acceptable

A1: $x = 25^\circ$

M1: For the key step in realising that $\cot x$ has a period of 180° and a second solution can be found by solving

$$x + 180^\circ = 3x - 50^\circ. \text{ The sight of } x = 115^\circ \text{ can imply this mark provided the step } x = 3x - 50^\circ \text{ has been seen.}$$

Using reciprocal functions it is for realising that $\tan x$ has a period of 180°

A1: $x = 115^\circ$ Withhold this mark if there are additional values in the range $(0, 180)$ but ignore values outside.

B1: Deduces that a solution can be found from $\cos x = 0 \Rightarrow x = 90^\circ$. Ignore additional values here.

Solutions with limited working. The question demands that candidates show all stages of working.

$$\text{SC: } \cos x \cot x = \cos x \cot(3x - 50^\circ) \Rightarrow \cot x = \cot(3x - 50^\circ) \Rightarrow x = 25^\circ, 115^\circ$$

They have shown some working so can score B1, B1 marked on open as 11000

Alt 1- Right hand side to left hand side

Question	Scheme	Marks	AOs
(a)	States or uses $\cot \theta = \frac{\cos \theta}{\sin \theta}$	B1	1.2
	$\cos \theta \cot \theta = \frac{\cos^2 \theta}{\sin \theta} = \frac{1 - \sin^2 \theta}{\sin \theta}$	M1	2.1
	$= \frac{1}{\sin \theta} - \sin \theta = \operatorname{cosec} \theta - \sin \theta \quad *$	A1*	2.1
		(3)	

Alt 2- Works on both sides

Question	Scheme	Marks	AOs
(a)	States or uses $\cot \theta = \frac{\cos \theta}{\sin \theta}$ or $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$	B1	1.2
	$LHS = \frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta} = \frac{\cos^2 \theta}{\sin \theta}$ $RHS = \cos \theta \cot \theta = \frac{\cos^2 \theta}{\sin \theta}$	M1	2.1
	States a conclusion E.g. "HENCE TRUE", "QED" or $\operatorname{cosec} \theta - \sin \theta = \cos \theta \cot \theta$ o.e. (condone = for $\equiv$)	A1*	2.1
		(3)	

Alt (b)

Question	Scheme	Marks	AOs
	$\cot x = \cot(3x - 50^\circ) \Rightarrow \frac{\cos x}{\sin x} = \frac{\cos(3x - 50^\circ)}{\sin(3x - 50^\circ)}$ $\sin(3x - 50^\circ)\cos x - \cos(3x - 50^\circ)\sin x = 0$ $\sin((3x - 50^\circ) - x) = 0$ $2x - 50^\circ = 0$	M1	3.1a
	$x = 25^\circ$	A1	1.1b
	Also $2x - 50^\circ = 180^\circ$	M1	2.1
	$x = 115^\circ$	A1	1.1b
	Deduces $\cos x = 0 \Rightarrow x = 90^\circ$	B1	2.2a
		(5)	

Q9.

Question	Scheme	Marks	AOs
(a)(i)	$y \times \frac{dx}{dt} = 5 \sin 2t \times 6 \cos t$ or $5 \times 2 \sin t \cos t \times 6 \cos t$	M1	1.2
	$(\text{Area} =) \int 5 \sin 2t \times 6 \cos t \, dt = \int 5 \times 2 \sin t \cos t \times 6 \cos t \, dt$ or $\int 5 \sin 2t \times 6 \cos t \, dt = \int 60 \sin t \cos^2 t \, dt$	dM1	1.1b
	$(\text{Area} =) \int_0^{\frac{\pi}{2}} 60 \sin t \cos^2 t \, dt *$	A1*	2.1*
	(3)		
(a)(ii)	$\int 60 \sin t \cos^2 t \, dt = -20 \cos^3 t$	M1 A1	1.1b 1.1b
	$\text{Area} = \left[-20 \cos^3 t \right]_0^{\frac{\pi}{2}} = 0 - (-20) = 20 *$	A1*	2.1
	(3)		
(b)	$5 \sin 2t = 4.2 \Rightarrow \sin 2t = \frac{4.2}{5}$	M1	3.4
	$t = 0.4986..., 1.072...$	A1	1.1b
	Attempts to find the x values at both t values	dM1	3.4
	$t = 0.4986... \Rightarrow x = 2.869...$ $t = 1.072 \Rightarrow x = 5.269...$	A1	1.1b
	Width of path = 2.40 metres	A1	3.2a
	(5)		
(11 marks)			

Notes:

(a)(i)

M1: Attempts to multiply y by $\frac{dx}{dt}$ to obtain $A \sin 2t \cos t$ but may apply $\sin 2t = 2 \sin t \cos t$ here

dM1: Attempts to use $\sin 2t = 2 \sin t \cos t$ within an integral which may be implied by

e.g. $\int \sin 2t \times \cos t \, dt = \int k \sin t \cos^2 t \, dt$

A1*: Fully correct work leading to the given answer.

This must include $\sin 2t = 2 \sin t \cos t$ or e.g. $5 \sin 2t = 10 \sin t \cos t$ seen explicitly in their proof and a correct intermediate line that includes an integral sign and the “ dt ”

Allow the limits to just “appear” in the final answer e.g. working need not be shown for the limits.

(a)(ii)

M1: Obtains $\int 60 \sin t \cos^2 t \, dt = k \cos^3 t$. This may be attempted via a substitution of $u = \cos t$ to obtain

$$\int 60 \sin t \cos^2 t \, dt = k u^3$$

A1: Correct integration $-20 \cos^3 t$ or equivalent e.g. $-20u^3$

A1*: Rigorous proof with all aspects correct including the correct limits and the $0 - (-20)$ and

not just: $-20 \cos^3 \frac{\pi}{2} - (-20 \cos^3 0) = 20$

(b)

M1: Uses the given model and attempts to find value(s) of t when $\sin 2t = \frac{4.2}{5}$. Look for $2t = \sin^{-1} \frac{4.2}{5} \Rightarrow t = \dots$

A1: At least one correct value for t , correct to 2 dp. FYI $t = 0.4986..., 1.072...$ or in degrees $t = 28.57..., 61.42...$

dM1: Attempts to find **TWO** distinct values of x when $\sin 2t = \frac{4.2}{5}$. Condone poor trig work and allow this mark if 2

values of x are attempted from 2 values of t .

A1: Both values correct to 2 dp. NB $x = 2.869\dots, 5.269\dots$

Or may take Cartesian approach

$$5 \sin 2t = 4.2 \Rightarrow 10 \sin t \cos t = 4.2 \Rightarrow 10 \frac{x}{6} \sqrt{1 - \frac{x^2}{36}} = 4.2 \Rightarrow x^4 - 36x^2 + 228.6144 = 0 \Rightarrow x = 2.869\dots, 5.269\dots$$

M1: For converting to Cartesian form **A1:** Correct quartic **M1:** Solves quartic **A1:** Correct values

A1: 2.40 metres or 240 cm

Allow awrt 2.40 m or allow 2.4m (not awrt 2.4 m) and allow awrt 240 cm. **Units are required.**

Q10.

Question	Scheme	Marks	AOs
(a)	Uses $\cos^2 x = 1 - \sin^2 x \Rightarrow 3 \sin^2 x + \sin x + 8 = 9(1 - \sin^2 x)$	M1	3.1a
	$\Rightarrow 12 \sin^2 x + \sin x - 1 = 0$	A1	1.1b
	$\Rightarrow (4 \sin x - 1)(3 \sin x + 1) = 0$	M1	1.1b
	$\Rightarrow \sin x = \frac{1}{4}, -\frac{1}{3}$	A1	1.1b
	Uses arcsin to obtain two correct values	M1	1.1b
	All four of $x = 14.48^\circ, 165.52^\circ, -19.47^\circ, -160.53^\circ$	A1	1.1b
		(6)	
(b)	Attempts $2\theta - 30^\circ = -19.47^\circ$	M1	3.1a
	$\Rightarrow \theta = 5.26^\circ$	A1ft	1.1b
		(2)	
(8 marks)			
Notes:			
(a)			
M1: Substitutes $\cos^2 x = 1 - \sin^2 x$ into $3 \sin^2 x + \sin x + 8 = 9 \cos^2 x$ to create a quadratic equation in just $\sin x$			
A1: $12 \sin^2 x + \sin x - 1 = 0$ or exact equivalent			
M1: Attempts to solve their quadratic equation in $\sin x$ by a suitable method. These could include factorisation, formula or completing the square.			
A1: $\sin x = \frac{1}{4}, -\frac{1}{3}$			
M1: Obtains two correct values for their $\sin x = k$			
A1: All four of $x = 14.48^\circ, 165.52^\circ, -19.47^\circ, -160.53^\circ$			
(b)			
M1: For setting $2\theta - 30^\circ = \text{their } -19.47^\circ$			
A1ft: $\theta = 5.26^\circ$ but allow a follow through on their -19.47°			

Q11.

Question	Scheme	Marks	AOs
(a)	$\frac{dH}{dt} = \frac{H \cos 0.25t}{40} \Rightarrow \int \frac{1}{H} dH = \int \frac{\cos 0.25t}{40} dt$	M1	3.1a
	$\ln H = \frac{1}{10} \sin 0.25t (+c)$	M1 A1	1.1b 1.1b
	Substitutes $t = 0, H = 5 \Rightarrow c = \ln(5)$	dM1	3.4
	$\ln\left(\frac{H}{5}\right) = \frac{1}{10} \sin 0.25t \Rightarrow H = 5e^{0.1 \sin 0.25t} *$	A1*	2.1
		(5)	
(b)	Max height = $5e^{0.1} = 5.53 \text{ m}$ (Condone lack of units)	B1	3.4
		(1)	
(c)	Sets $0.25t = \frac{5\pi}{2}$	M1	3.1b
	31.4	A1	1.1b
		(2)	
(8 marks)			

(a)

M1: Separates the variables to reach $\int \frac{1}{H} dH = \int \frac{\cos 0.25t}{40} dt$ or equivalent.

The integral signs need to be present on both sides and the dH AND dt need to be in the correct positions.

M1: Integrates both sides to reach $\ln H = A \sin 0.25t$ or equivalent with or without the $+c$

A1: $\ln H = \frac{1}{10} \sin 0.25t + c$ or equivalent with or without the $+c$. Allow two constants, one either side

If the 40 was on the lhs look for $40 \ln H = 4 \sin 0.25t + c$ or equivalent.

dM1: Substitutes $t = 0, H = 5 \Rightarrow c = \dots$ There needs to have been a single $+c$ to find.

It is dependent upon the previous M mark. You may allow even if you don't explicitly see $t = 0, H = 5$ as it may be implied from their previous line

If the candidate has attempted to change the subject and made an error/ slip then condone it for this M but not the final A. Eg. $40 \ln H = 4 \sin 0.25t + c \Rightarrow H^{40} = e^{4 \sin 0.25t} + e^c \Rightarrow 5^{40} = 1 + e^c \Rightarrow c = \dots$

Also many students will be attempting to get to the given answer so condone the method of finding $c = \dots$
These students will lose the A1* mark

A1*: Proceeds via $\ln H = \frac{1}{10} \sin 0.25t + \ln 5$ or equivalent to the given answer $H = 5e^{0.1 \sin 0.25t}$ with at least one correct intermediate line and no incorrect work.

DO NOT condone c 's going to e^c 's when they should be e^c or A

Accept as a minimum $\ln H = \frac{1}{10} \sin 0.25t + \ln 5 \Rightarrow H = e^{\frac{1}{10} \sin 0.25t + \ln 5}$ or $H = e^{\frac{1}{10} \sin 0.25t} \times e^{\ln 5}$ before sight of the given answer

If the only error was to omit the integration signs on line 1, thus losing the first M1, allow the candidate to have access to this mark following a correct intermediate line (see above).

If they attempt to change the subject first then the constant of integration must have been adapted if the A1* is to be awarded. $\ln H = \frac{1}{10} \sin 0.25t + c \Rightarrow H = e^{\frac{1}{10} \sin 0.25t + c} \Rightarrow H = Ae^{\frac{1}{10} \sin 0.25t}$

The dM1 and A1* under this method are awarded at virtually the same time.

Also, for the final two marks, you may see a proof from $\int_0^H \frac{40}{H} dH = \int_5^t \cos 0.25t dt$

.....
There is an alternative via the use of an integrating factor.

(b)

B1: States that the maximum height is 5.53 m Accept $5e^{0.1}$ Condone a lack of units here, but penalise if incorrect units are used.

(c)

M1: For identifying that it would reach the maximum height for the 2nd time when $0.25t = \frac{5\pi}{2}$ or 450

A1: Accept awrt 31.4 or 10π Allow if units are seen

Q12.

Question	Scheme	Marks	AOs
	(i) $4\sin x = \sec x$, $0 \leq x < \frac{\pi}{2}$; (ii) $5\sin \theta - 5\cos \theta = 2$, $0 \leq \theta < 360^\circ$		
(i) Way 1	For $\sec x = \frac{1}{\cos x}$	B1	1.2
	$\{4\sin x = \sec x \Rightarrow 4\sin x \cos x = 1 \Rightarrow 2\sin 2x = 1 \Rightarrow \sin 2x = \frac{1}{2}\}$	M1	3.1a
	$x = \frac{1}{2}\arcsin\left(\frac{1}{2}\right)$ or $\frac{1}{2}\left(\pi - \arcsin\left(\frac{1}{2}\right)\right) \Rightarrow x = \frac{\pi}{12}, \frac{5\pi}{12}$	dM1	1.1b
		A1	1.1b
	(4)		
(i) Way 2	For $\sec x = \frac{1}{\cos x}$	B1	1.2
	$\{4\sin x = \sec x \Rightarrow 4\sin x \cos x = 1 \Rightarrow 16\sin^2 x \cos^2 x = 1$ $16\sin^2 x(1 - \sin^2 x) = 1$ $16(1 - \cos^2 x)\cos^2 x = 1$ $16\sin^4 x - 16\sin^2 x + 1 = 0$ $16\cos^4 x - 16\cos^2 x + 1 = 0$ $\sin^2 x$ or $\cos^2 x = \frac{16 \pm \sqrt{192}}{32} = \frac{2 \pm \sqrt{3}}{4}$ or $0.933\dots, 0.066\dots$	M1	3.1a
	$x = \arcsin\left(\sqrt{\frac{2 \pm \sqrt{3}}{4}}\right)$ or $x = \arccos\left(\sqrt{\frac{2 \pm \sqrt{3}}{4}}\right) \Rightarrow x = \frac{\pi}{12}, \frac{5\pi}{12}$	dM1	1.1b
		A1	1.1b
	(4)		

(ii)	Complete strategy, i.e.			
	- Expresses $5\sin\theta - 5\cos\theta = 2$ in the form $R\sin(\theta - \alpha) = 2$, finds both R and α, and proceeds to $\sin(\theta - \alpha) = k$, $k < 1$, $k \neq 0$ - Applies $(5\sin\theta - 5\cos\theta)^2 = 2^2$, followed by applying both $\cos^2\theta + \sin^2\theta = 1$ and $\sin 2\theta = 2\sin\theta\cos\theta$ to proceed to $\sin 2\theta = k$, $k < 1$, $k \neq 0$		M1	3.1a
	$R = \sqrt{50}$ $\tan\alpha = 1 \Rightarrow \alpha = 45^\circ$	$(5\sin\theta - 5\cos\theta)^2 = 2^2 \Rightarrow$ $25\sin^2\theta + 25\cos^2\theta - 50\sin\theta\cos\theta = 4$ $\Rightarrow 25 - 25\sin 2\theta = 4$	M1	1.1b
	$\sin(\theta - 45^\circ) = \frac{2}{\sqrt{50}}$	$\sin 2\theta = \frac{21}{25}$	A1	1.1b
	dependent on the first M mark			
	e.g. $\theta = \arcsin\left(\frac{2}{\sqrt{50}}\right) + 45^\circ$	e.g. $\theta = \frac{1}{2}\left(\arcsin\left(\frac{21}{25}\right)\right)$	dM1	1.1b
	$\theta = \text{awrt } 61.4^\circ, \text{ awrt } 208.6^\circ$		A1	2.1
Note: Working in radians does not affect any of the first 4 marks				
			(5)	
(9 marks)				

Question	Scheme	Marks	AOs
	(ii) $5\sin\theta - 5\cos\theta = 2, 0 \leq \theta < 360^\circ$		
(ii) Alt 1	Complete strategy, i.e. Attempts to apply $(5\sin\theta)^2 = (2+5\cos\theta)^2$ or $(5\sin\theta - 2)^2 = (5\cos\theta)^2$ followed by applying $\cos^2\theta + \sin^2\theta = 1$ and solving a quadratic equation in either $\sin\theta$ or $\cos\theta$ to give at least one of $\sin\theta = k$ or $\cos\theta = k, k < 1, k \neq 0$	M1	3.1a
	e.g. $25\sin^2\theta = 4 + 20\cos\theta + 25\cos^2\theta$ $\Rightarrow 25(1 - \cos^2\theta) = 4 + 20\cos\theta + 25\cos^2\theta$ or e.g. $25\sin^2\theta - 20\sin\theta + 4 = 25\cos^2\theta$ $\Rightarrow 25\sin^2\theta - 20\sin\theta + 4 = 25(1 - \sin^2\theta)$	M1	1.1b
	$50\cos^2\theta + 20\cos\theta - 21 = 0$	$50\sin^2\theta - 20\sin\theta - 21 = 0$	
	$\cos\theta = \frac{-20 \pm \sqrt{4600}}{100}, \text{ o.e.}$	$\sin\theta = \frac{20 \pm \sqrt{4600}}{100}, \text{ o.e.}$	A1
	dependent on the first M mark e.g. $\theta = \arccos\left(\frac{-2 + \sqrt{46}}{10}\right)$		dM1
	e.g. $\theta = \arcsin\left(\frac{2 + \sqrt{46}}{10}\right)$		
	$\theta = \text{awrt } 61.4^\circ, \text{ awrt } 208.6^\circ$		A1
		(5)	
Notes for Question			
(i)			
B1:	For recalling that $\sec x = \frac{1}{\cos x}$		
M1:	Correct strategy of - Way 1: applying $\sin 2x = 2\sin x \cos x$ and proceeding to $\sin 2x = k, k \leq 1, k \neq 0$ - Way 2: squaring both sides, applying $\cos^2 x + \sin^2 x = 1$ and solving a quadratic equation in either $\sin^2 x$ or $\cos^2 x$ to give $\sin^2 x = k$ or $\cos^2 x = k, k \leq 1, k \neq 0$		
dM1:	Uses the correct order of operations to find at least one value for x in either radians or degrees		
A1:	Clear reasoning to achieve both $x = \frac{\pi}{12}, \frac{5\pi}{12}$ and no other values in the range $0 \leq x < \frac{\pi}{2}$		
Note:	Give dM1 for $\sin 2x = \frac{1}{2} \Rightarrow$ any of $\frac{\pi}{12}, \frac{5\pi}{12}, 15^\circ, 75^\circ, \text{ awrt } 0.26 \text{ or awrt } 1.3$		
Note:	Give special case, SC B1M0M0A0 for writing down any of $\frac{\pi}{12}, \frac{5\pi}{12}, 15^\circ \text{ or } 75^\circ$ with no working		

Notes for Question Continued	
(ii)	
M1:	See scheme
Note:	Alternative strategy: Expresses $5\sin\theta - 5\cos\theta = 2$ in the form $R\cos(\theta + \alpha) = -2$, finds both R and α , and proceeds to $\cos(\theta + \alpha) = k$, $ k < 1$, $k \neq 0$
M1:	Either • uses $R\sin(\theta - \alpha)$ to find the values of both R and α • attempts to apply $(5\sin\theta - 5\cos\theta)^2 = 2^2$, uses $\cos^2\theta + \sin^2\theta = 1$ and proceeds to find an equation of the form $\pm\lambda \pm \mu\sin 2\theta = \pm\beta$ or $\pm\mu\sin 2\theta = \pm\beta$; $\mu \neq 0$ • attempts to apply $(5\sin\theta)^2 = (2 + 5\cos\theta)^2$ or $(5\sin\theta - 2)^2 = (5\cos\theta)^2$ and uses $\cos^2\theta + \sin^2\theta = 1$ to form an equation in $\cos\theta$ only or $\sin\theta$ only
A1:	For $\sin(\theta - 45^\circ) = \frac{2}{\sqrt{50}}$, o.e., $\cos(\theta + 45^\circ) = -\frac{2}{\sqrt{50}}$, o.e. or $\sin 2\theta = \frac{21}{25}$, o.e. or $\cos\theta = \frac{-20 \pm \sqrt{4600}}{100}$, o.e. or $\cos\theta = \text{awrt } 0.48$, awrt -0.88 or $\sin\theta = \frac{20 \pm \sqrt{4600}}{100}$, o.e., or $\sin\theta = \text{awrt } 0.88$, awrt -0.48
Note:	$\sin(\theta - 45^\circ)$, $\cos(\theta + 45^\circ)$, $\sin 2\theta$ must be made the subject for A1
dM1:	dependent on the first M mark Uses the correct order of operations to find at least one value for x in either degrees or radians
Note:	dM1 can also be given for $\theta = 180^\circ - \arcsin\left(\frac{2}{\sqrt{50}}\right) + 45^\circ$ or $\theta = \frac{1}{2}\left(180^\circ - \arcsin\left(\frac{21}{25}\right)\right)$
A1:	Clear reasoning to achieve both $\theta = \text{awrt } 61.4^\circ$, awrt 208.6° and no other values in the range $0 \leq \theta < 360^\circ$
Note:	Give M0M0A0M0A0 for writing down any of $\theta = \text{awrt } 61.4^\circ$, awrt 208.6° with no working
Note:	Alternative solutions: (to be marked in the same way as Alt 1): • $5\sin\theta - 5\cos\theta = 2 \Rightarrow 5\tan\theta - 5 = 2\sec\theta \Rightarrow (5\tan\theta - 5)^2 = (2\sec\theta)^2$ $\Rightarrow 25\tan^2\theta - 50\tan\theta + 25 = 4\sec^2\theta \Rightarrow 25\tan^2\theta - 50\tan\theta + 25 = 4(1 + \tan^2\theta)$ $\Rightarrow 21\tan^2\theta - 50\tan\theta + 21 = 0 \Rightarrow \tan\theta = \frac{50 \pm \sqrt{736}}{42} = \frac{25 \pm 2\sqrt{46}}{21} = 1.8364\dots, 0.5445\dots$ $\Rightarrow \theta = \text{awrt } 61.4^\circ$, awrt 208.6° only • $5\sin\theta - 5\cos\theta = 2 \Rightarrow 5 - 5\cot\theta = 2\text{cosec}\theta \Rightarrow (5 - 5\cot\theta)^2 = (2\text{cosec}\theta)^2$ $\Rightarrow 25 - 50\cot\theta + 25\cot^2\theta = 4\text{cosec}^2\theta \Rightarrow 25 - 50\cot\theta + 25\cot^2\theta = 4(1 + \cot^2\theta)$ $\Rightarrow 21\cot^2\theta - 50\cot\theta + 21 = 0 \Rightarrow \cot\theta = \frac{50 \pm \sqrt{736}}{42} = \frac{25 \pm 2\sqrt{46}}{21} = 1.8364\dots, 0.5445\dots$ $\Rightarrow \theta = \text{awrt } 61.4^\circ$, awrt 208.6° only

Q13.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$5 \sin 2\theta = 9 \tan \theta \Rightarrow$ $10 \sin \theta \cos \theta = 9 \times \frac{\sin \theta}{\cos \theta}$	M1	This mark is given for a method to substitute terms to form an equation in terms of $\cos \theta$
	$10 \cos^2 \theta = 9$	M1	This mark is given for a correct equation in terms of $\cos \theta$
	$\theta = \arccos \pm \sqrt{\frac{9}{10}}$	M1	This mark is given for finding a value for θ in terms of arccos
	$\theta = \pm 18.4^\circ, \pm 161.6^\circ$	A1	This mark is given for any one value of 18.4° or 161.6° found.
		A1	This mark is given for four values of θ found correctly
	$\theta = \pm 0^\circ, 180^\circ$	B1	This mark is given for the deduction of the two other solutions for θ
(b)	$10 \cos^2 (x - 25) = 9$ x has smallest positive value when $x - 25^\circ = -18.4^\circ$	M1	This mark is given for finding an equation to solve for x
	$x = 6.6^\circ$	A1	This mark is given for correctly finding the smallest positive solution to the equation

Q14.

Question	Scheme	Marks	AOs
(a)	$R = \sqrt{5}$	B1	1.1b
	$\tan \alpha = 2 \Rightarrow \alpha = \dots$	M1	1.1b
	$\alpha = 1.107$	A1	1.1b
		(3)	
	$\theta = 5 + \sqrt{5} \sin \left(\frac{\pi t}{12} + 1.107 - 3 \right)$		
(b)	$(5 + \sqrt{5})^\circ \text{C}$ or awrt 7.24°C	B1ft	2.2a
		(1)	
(c)	$\frac{\pi t}{12} + 1.107 - 3 = \frac{\pi}{2} \Rightarrow t =$	M1	3.1b
	$t =$ awrt 13.2	A1	1.1b
	Either 13:14 or 1:14 pm or 13 hours 14 minutes after midnight.	A1	3.2a
		(3)	
(7 marks)			
Notes:			

(a)

B1: $R = \sqrt{5}$ only.

M1: Proceeds to a value of α from $\tan \alpha = \pm 2$, $\tan \alpha = \pm \frac{1}{2}$, $\sin \alpha = \pm \frac{2}{\sqrt{5}}$ OR $\cos \alpha = \pm \frac{1}{\sqrt{5}}$

It is implied by either awrt 1.11 (radians) or 63.4 (degrees)

A1: $\alpha = \text{awrt } 1.107$

(b)

B1ft: Deduces that the maximum temperature is $(5 + \sqrt{5})^\circ\text{C}$ or awrt 7.24°C Remember to isw
Condone a lack of units. Follow through on their value of R so allow $(5 + R)^\circ\text{C}$

(c)

M1: An complete strategy to find t from $\frac{\pi t}{12} \pm 1.107 - 3 = \frac{\pi}{2}$.

Follow through on their 1.107 but the angle must be in radians.

It is possible via degrees but only using $15t \pm 63.4 - 171.9 = 90$

A1: awrt $t = 13.2$

A1: The question asks for the time of day so accept either 13:14, 1:14 pm, 13 hours 14 minutes after midnight, 13h 14, or 1 hour 14 minutes after midday. If in doubt use review

It is possible to attempt parts (b) and (c) via differentiation but it is unlikely to yield correct results.

$$\frac{d\theta}{dt} = \frac{\pi}{12} \cos\left(\frac{\pi t}{12} - 3\right) - \frac{2\pi}{12} \sin\left(\frac{\pi t}{12} - 3\right) = 0 \Rightarrow \tan\left(\frac{\pi t}{12} - 3\right) = \frac{1}{2} \Rightarrow t = 13.23 = 13:14 \text{ scores M1 A1 A1}$$

$$\frac{d\theta}{dt} = \cos\left(\frac{\pi t}{12} - 3\right) - 2 \sin\left(\frac{\pi t}{12} - 3\right) = 0 \Rightarrow \tan\left(\frac{\pi t}{12} - 3\right) = \frac{1}{2} \Rightarrow t = 13.23 = 13:14 \text{ they can score M1 A0 A1 (SC)}$$

A value of $t = 1.23$ implies the minimum value has been found and therefore incorrect method M0.

Q15.

Question	Scheme	Marks	AOs
(a)	Attempts to differentiate using the quotient rule or otherwise	M1	2.1
	$f'(x) = \frac{e^{\sqrt{2}x-1} \times 8 \cos 2x - 4 \sin 2x \times \sqrt{2} e^{\sqrt{2}x-1}}{(e^{\sqrt{2}x-1})^2}$	A1	1.1b
	Sets $f'(x) = 0$ and divides/ factorises out the $e^{\sqrt{2}x-1}$ terms	M1	2.1
	Proceeds via $\frac{\sin 2x}{\cos 2x} = \frac{8}{4\sqrt{2}}$ to $\Rightarrow \tan 2x = \sqrt{2}^*$	A1*	1.1b
		(4)	
(b)	(i) Solves $\tan 4x = \sqrt{2}$ and attempts to find the 2 nd solution	M1	3.1a
	$x = 1.02$	A1	1.1b
	(ii) Solves $\tan 2x = \sqrt{2}$ and attempts to find the 1 st solution	M1	3.1a
	$x = 0.478$	A1	1.1b
		(4)	
(8 marks)			

Notes:

(a)

M1: Attempts to differentiate by using the quotient rule with $u = 4 \sin 2x$ and $v = e^{\sqrt{2}x-1}$ or alternatively uses the product rule with $u = 4 \sin 2x$ and $v = e^{1-\sqrt{2}x}$

A1: For achieving a correct $f'(x)$. For the product rule

$$f'(x) = e^{1-\sqrt{2}x} \times 8 \cos 2x + 4 \sin 2x \times -\sqrt{2}e^{1-\sqrt{2}x}$$

M1: This is scored for cancelling/ factorising out the exponential term. Look for an equation in just $\cos 2x$ and $\sin 2x$

A1*: Proceeds to $\tan 2x = \sqrt{2}$. This is a given answer.

(b) (i)

M1: Solves $\tan 4x = \sqrt{2}$ attempts to find the 2nd solution. Look for $x = \frac{\pi + \arctan \sqrt{2}}{4}$

Alternatively finds the 2nd solution of $\tan 2x = \sqrt{2}$ and attempts to divide by 2

A1: Allow awrt $x = 1.02$. The correct answer, with no incorrect working scores both marks

(b)(ii)

M1: Solves $\tan 2x = \sqrt{2}$ attempts to find the 1st solution. Look for $x = \frac{\arctan \sqrt{2}}{2}$

A1: Allow awrt $x = 0.478$. The correct answer, with no incorrect working scores both marks